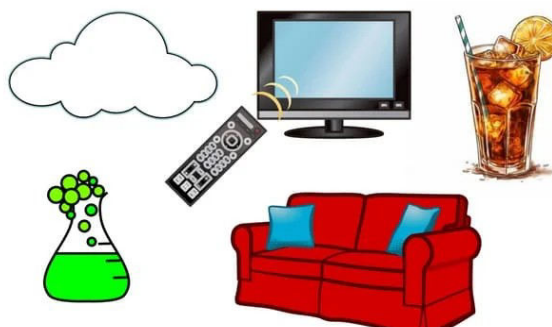




MATTER IN OUR SURROUNDINGS

- x Everything in this universe is made up of material which scientists have named “MATTER”.
- x **Matter is defined as anything that occupy space and have mass.**
- x The air we breathe, the food we eat, stones, clouds, stars, plants and animals, even a small drop of water or a particle of sand – everything is matter.
- x Early Indian philosophers classified matter in the form of five basic elements – the “**Panch Tatva**” – *air, earth, fire, sky and water*.
- x Modern day scientists have evolved two types of classification of matter based on their physical and chemical properties.



PHYSICAL NATURE OF MATTER

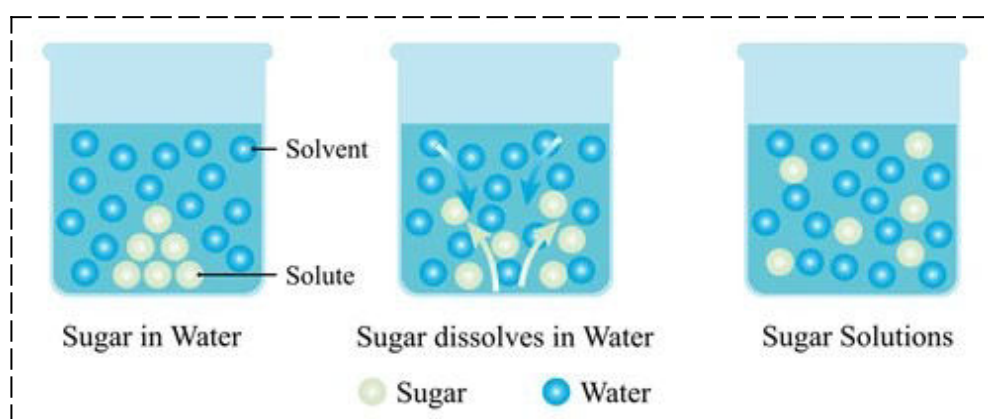
- x Matter is made up of millions of tiny particles.
- x The particle of one type of matter is different from the particle of another type of matter.

The particles of matter are very small – they are small beyond our imagination!

CHARACTERISTICS OF PARTICLES OF MATTER

1. Particles of matter have space between them.

For Example, *the formation of sugar solution* – when we make sugar solution, the particles of one type of matter get into the space between the particles of the other. This shows that there is enough space between particles of matter.



2. Particles of matter are continuously moving.

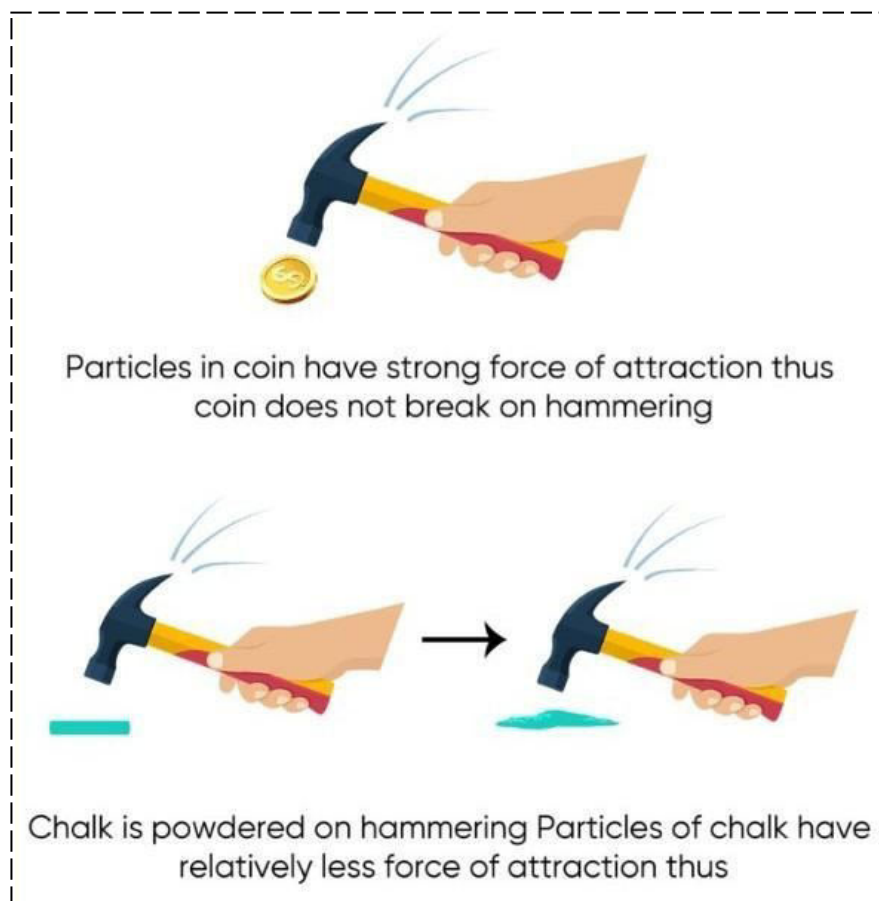
For example, *mixing of ink in water* – Particles of matter are continuously moving and possess kinetic energy. As the temperature rises, particles move faster. This means that with an increase in temperature the kinetic energy of the particles also increases.

- Intermixing the particles of two different types of matter on their own is called **diffusion**.
- On heating, diffusion becomes faster.



3. Particles of matter attract each other.

For example, Each matter have force of attraction between the particles which varies depending on the objects.



Question:

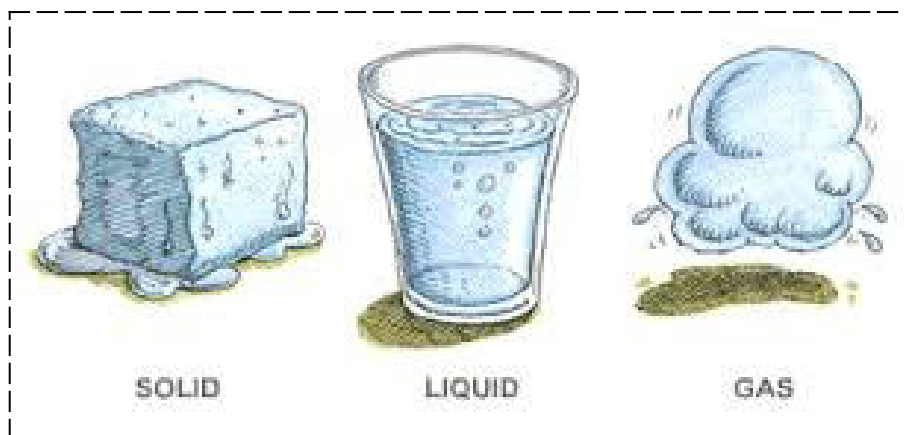
A diver is able to cut through water in a swimming pool. Which property of matter does this observation show?

Answer:

This shows that the particles of water has space between them and have weak intermolecular force of attraction between them.

STATES OF MATTER

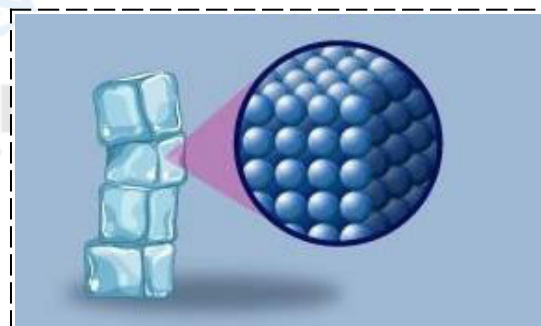
- ✕ Matter around us exists in three different states – **Solid, Liquid and Gas.**



- ✕ These states of matter arise due to the variation in the characteristics of the particles of matter.

THE SOLID STATE

The particles of solid are very close to each other. They have negligible space between them and they vibrate at their place.



Properties Of Solid

- ✕ The particles of solid have a strong intermolecular force of attraction between them.
- ✕ The solid has a fixed shape and volume.
- ✕ Solids have tendency to maintain their shape when subjected to outside force.
- ✕ The solid has a rigid (may break under force but difficult to change their shape) structure.
- ✕ The solids are incompressible.

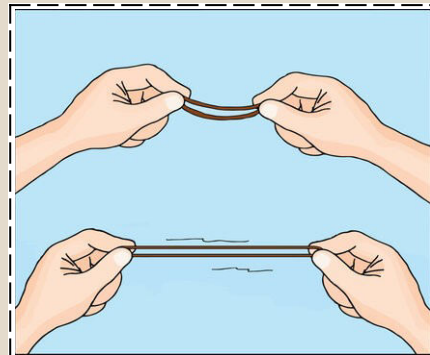
SPECIAL CASES

Question 1:

What about a rubber band, can it change its shape on stretching? Is it a solid?

Answer:

Rubber is a solid as rubber band changes shape under force and regains the same shape when the force is removed. If excessive force is applied, it breaks.



Question 2:

What about sugar and salt? When kept in different jars these take the shape of the jar. Are they solid?

Answer:

Sugar and salt are solid as the shape of each individual sugar or salt crystal remains fixed, whether we take it in our hand, put it in a plate or in a jar.



Question 3:

What about a sponge? It is a solid yet we are able to compress it. Why?

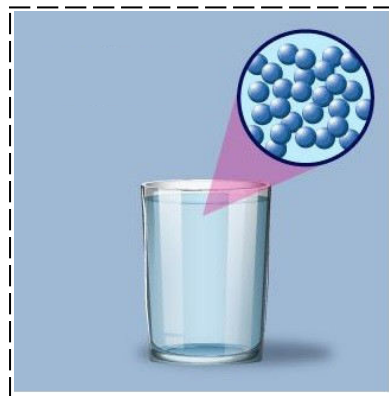
Answer:

Sponge is a solid also and it has minute holes, in which air is trapped, when we press it, the air is expelled out and we are able to compress it.



THE LIQUID STATE

The particles of liquids are slightly far from each other so the force of attraction between the particles are not so strong and particles can freely move.



Properties Of Liquid

- ✕ Liquids have no fixed shape but have a fixed volume. They take the shape of the container in which they are kept.
- ✕ The particles of liquid have intermediate intermolecular force of attraction between them.
- ✕ Liquids flow and change shape, so they are not rigid but can be called fluid.
- ✕ Liquids are incompressible.
- ✕ The rate of diffusion of liquids is higher than that of solid due to the fact that in the liquid state, particles move freely and have greater space between each other as compared to the particles of solid.

Question:

Liquids generally have a lower density as compared to solids. But you must have observed that ice floats on water. Find out why.

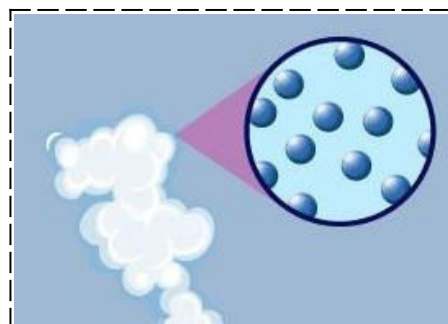
Answer:

The ice is solid, but it has a low density than water because of the cage like structure of ice. It has lot of spaces between them, so it floats on water.



THE GASEOUS STATE

The particles of gas are very far from each other. So, the force of attraction between the particles are weak and particles can move freely in all directions.



Properties Of Gas

- x The particles of gas have weak intermolecular force of attraction between them.
- x The gas has neither fixed shape nor fixed volume.
- x The gas can take the shape of the container.
- x The gas has no rigid structure.
- x Gas is highly compressible as compared to solids and liquids.
Example – CNG (Compressed Natural Gas) is used as fuel in vehicles, due to high compressibility of gas, large volume of gas can be compressed into a small cylinder and transported easily.
- x Gas can flow easily.

Distinguishing properties of solids, liquids and gases

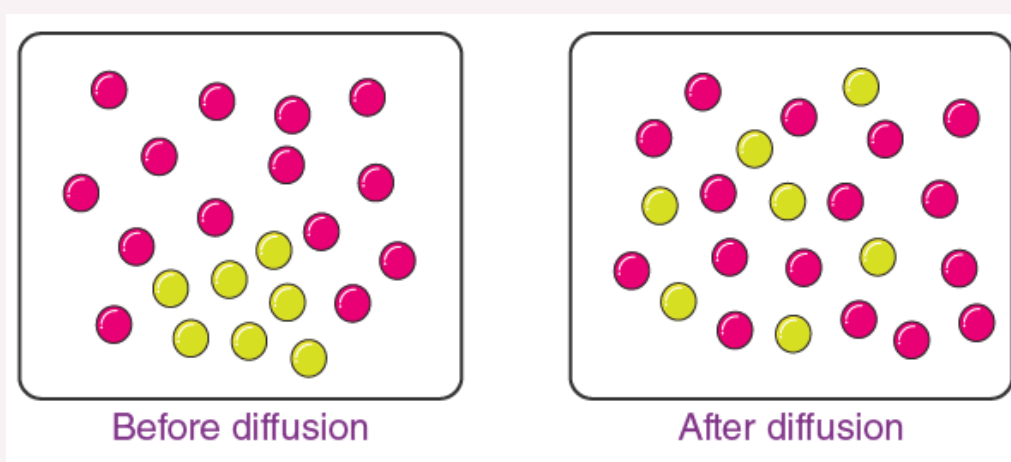
	SOLIDS	LIQUIDS	GASES
Volume	The solids have a definite volume	The liquids have a definite volume	The gases occupy the available volume
Shape	The solids have a definite shape	The liquids do not have a definite shape	The gases do not have a definite shape
Fluidity	The solids cannot flow.	The liquids can flow	The gases can flow
Rigidity	Highly rigid	Less rigid	Not rigid
Compressibility	Cannot be compressed	Negligibly compressed	Highly compressible

DIFFUSION

- The process of **intermixing of particles of two different types** of matter is called diffusion.
- The order for the **rate of diffusion** in different state of matter:

GAS > LIQUID > SOLID

- The rate of diffusion increases with an increase in temperature because when we increase the temperature, the kinetic energy of the particles increases and they mix into other quickly.



Question:

Arrange the following substances in increasing order of forces of attraction between the particles – water, sugar, oxygen.

Answer:

The force of attraction is strong in solid and weak in gas.

Sugar is solid, water is liquid and oxygen is gas. So, the order of force of attraction between the particles is:

OXYGEN < WATER < SUGAR

CAN MATTER CHANGE ITS STATE?

Water exists in all three states of matter and **these states of matter can be interconverted into each other.**



SAATVIK
STUDY STATION

A) Melting and Fusion -

The process in which the substances changes its state from **solid to liquid** is called Melting and fusion. The temperature at which the solid starts converting into a liquid at atmospheric pressure is called **Melting point**.

The **melting point of ice is 273.15 K** (kelvin i.e. SI unit of temperature, $0^{\circ}\text{C} = 273.15\text{ K}$)

Example – Melting of Ice-cream



B) Vaporization -

The process in which the substances change its state from **liquid to gas** is called vaporization. The temperature at which the liquid starts converting into a gas at atmospheric pressure is called the **boiling point**.

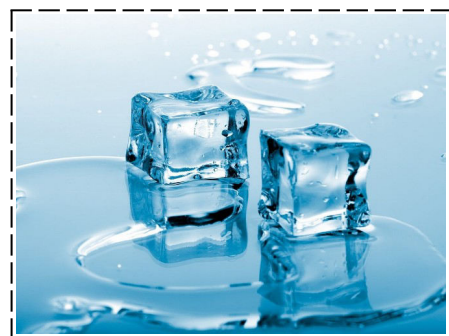
Example – Formation of steam.



C) Freezing and Solidification -

The process in which the substance changes its state from **liquid to solid** is called solidification or freezing.

Example – Formation of Ice.



D) Sublimation -

The process in which the substance changes its state from **solid to gas** is called sublimation.

Example – Sublimation of dry ice.



E) Deposition -

The process in which the substance changes its state from **gas to solid** is called deposition.

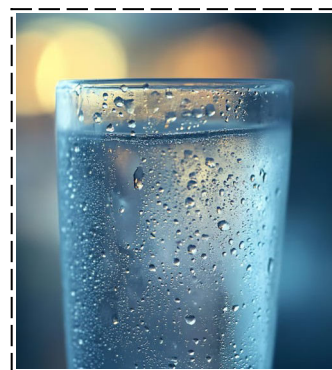
Example – Formation of frost on the leaves.



F) Condensation -

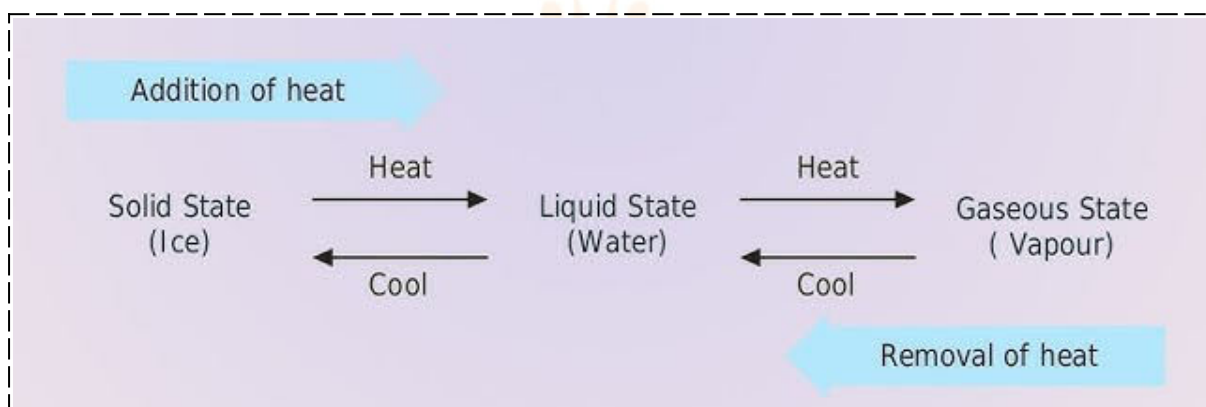
The process in which the substance changes its state from **gas to liquid** is called condensation.

Example – The formation of water droplets on the cold surface.



EFFECT OF CHANGE OF TEMPERATURE

- When the temperature of the matter is increased, the kinetic energy of the particles of matter also increases. Due to the increase in kinetic energy, the particles of matter start vibrating with greater speed and move freely.



- When the temperature of the matter is decreased, the kinetic energy of the particles of matter also decreases. Due to the decrease in kinetic energy, the particles of matter start coming closer and developing stronger intermolecular attraction. Hence, the particles restrict their motion.
- The melting point of ice is 273.15 K ($0^{\circ}\text{C} = 273.15\text{ K}$)**

The relation of the Kelvin and Celsius scales is -

$$T(\text{K}) = T(^{\circ}\text{C}) + 273.15$$

Question:

***Very Important**

When a solid melts, its temperature remains the same, so where does the heat energy go?

OR

Explain Latent heat of fusion?

Answer:

During the process of melting, the temperature of the system does not change after the melting point is reached, till all the ice melts. This happens even though we continue to heat the beaker, i.e. we continue to supply heat.

This **heat gets used up in changing the state** by overcoming the forces of attraction between the particles. As the heat is absorbed by the ice without showing any rise in the temperature, it is considered that it gets hidden in the contents of the beaker and is known as the **latent heat** or hidden heat.

The amount of heat energy required to change 1 Kg of a solid into liquid at atmospheric pressure at its melting point is known as **latent heat of fusion**.

So, the particles in water at 0°C (273.15 K) have more energy as compared to particles of ice at same temperature.

- Also, when we supply heat to water, particles start moving even faster. At a certain temperature, a point is reached when the particles have enough energy to break free from the force of attraction. **At this temperature the liquid starts changing into gas.**
- The temperature at which liquid starts boiling at atmospheric pressure is known as **boiling point**.
- Particles from the bulk of liquids gain enough energy to change into vapour state.
- **The boiling point of water is 373 K or 100°C**
- The amount of heat required to change 1 mole of liquid into a gas at atmospheric pressure at its boiling point is called **latent heat of vaporisation**.
- Particles in steam, i.e., **water vapours at 373 K have more energy than water at the same temperature** because particles in steam have absorbed extra energy in the form of latent heat of vaporisation.

Question:

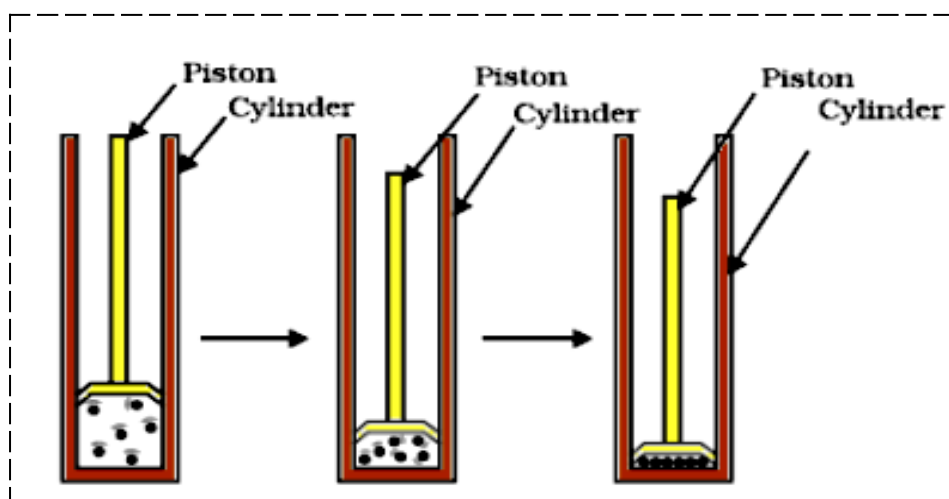
Why steam cause severe burn than hot water?

Answer:

The steam has more heat than water which is latent heat of vaporisation. That is why steam burns more than hot water.

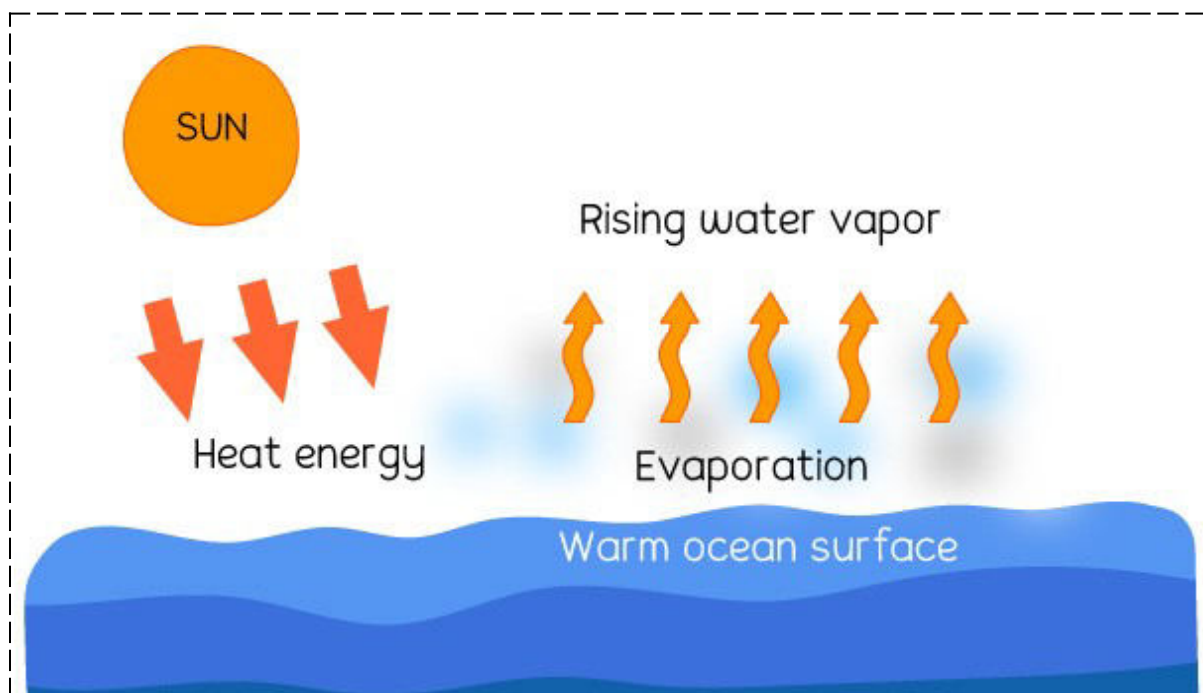
EFFECTS OF CHANGE IN PRESSURE

- x The difference in various state of matter is due to the difference in the distance between the constituent particles.
- x The gases have the most space between their particles and solid have the least. So, **solids and liquids are incompressible** and have the negligible effect of pressure change on their change in state.
- x When the pressure of the gas is increased, the kinetic energy of the molecules decreases and the speed of the molecules also decreases. So, the molecule starts coming closer to each other and developing a stronger force of attraction between them.
- x Applying pressure and reducing temperature can liquefy gases.
- x **Solid CO₂** is stored under high pressure. Solid CO₂ gets directly converted into gaseous state on decreasing the pressure by 1 atmosphere (*atm; unit of measuring pressure exerted by a gas, SI unit of pressure is Pascal (Pa)*) without coming into liquid state. This is the reason why solid carbon dioxide is also known as **dry ice**.



EVAPORATION

- In liquids, a small fraction of particles at the surface, having higher kinetic energy, is able to break away from the force of attraction of other particles and get converted into vapours.
- **The phenomenon of change of liquid into vapours at any temperature below its boiling point is called Evaporation.**
Example – Water when left uncovered slowly change into vapour.
Wet clothes dry up.



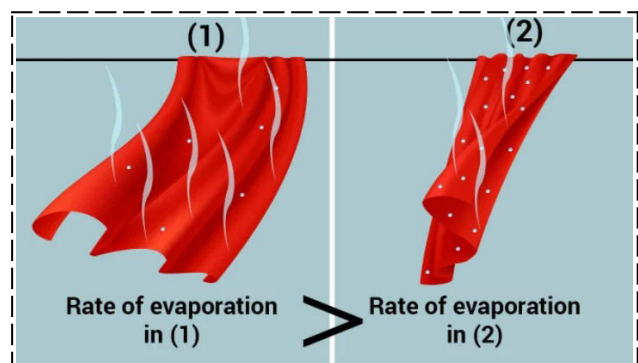
FACTORS AFFECTING EVAPORATION

The rate of evaporation is affected by -

1) Increase of surface area -

As evaporation is a surface phenomenon, if the surface area is increased, the rate of evaporation increases.

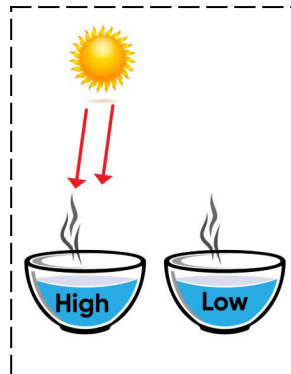
Example – while putting clothes for drying up we spread them out.



2) Increase in temperature -

With the increase of temperature, more number of particles get enough kinetic energy to go into the vapour state.

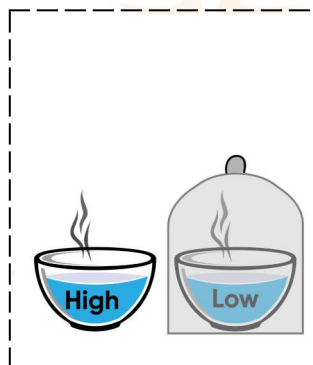
Example- Wet clothes dry quickly in hot summer than in winter.



3) Decrease in humidity -

When the amount of humidity (water vapours) in air is high, the rate of evaporation decreases.

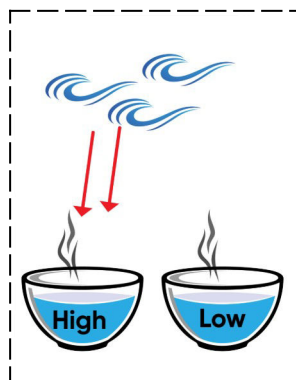
Example – Clothes dry up late on a rainy day due to high humidity in atmosphere.



4) Increase in wind speed -

The increase in wind speed will increase the rate of evaporation.

Example – Wet hair dries quickly when we blow dry it.



COOLING EFFECT OF EVAPORATION

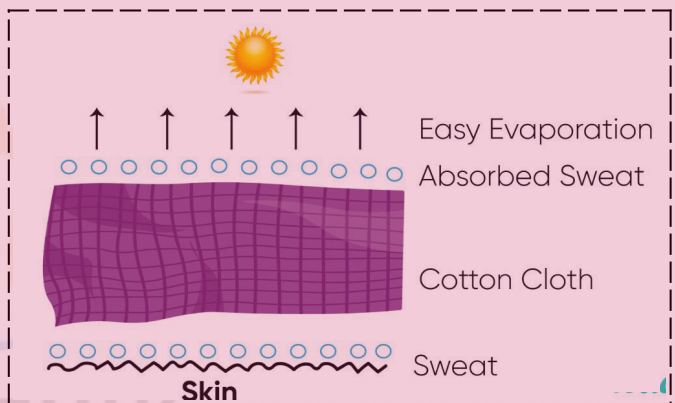
- ✗ In an open vessel, the liquid keeps on evaporating. The particles of liquid absorb energy from the surrounding to regain the energy lost during evaporation. This absorption of energy from the surroundings makes the surrounding cold.
- ✗ Example – After a hot sunny day, people sprinkle water in the room or open ground because a large amount of latent heat of vaporisation of water helps to cool the hot surface.

Question:

Why we should wear cotton clothes in summer?

Answer:

During evaporation, the particles at the surface of the liquid gain energy from the surroundings or body surface and change into vapours. The heat energy equal to the latent heat of vaporisation is absorbed from the body, leaving the body cool. Cotton being a good absorber of water helps in absorbing the sweat and exposing it to the atmosphere for easy evaporation.



Question:

Explain why does a desert cooler cool better on a hot dry day?

Answer:

The outer wall of the cooler is sprinkled by water constantly, which evaporates due to hot dry weather. This causes the cooling of air which is inside the cooler. The fan sends these cool air inside the room.

